ie UNIVERSITY | SCHOOL OF SCIENCE & TECHNOLOGY

BACHELOR IN DATA AND BUSINESS ANALYTICS

TIME SERIES ANALYSIS

Session 3 · Lags, Autocorrelation and the ACF

Professor: Angel Mora Segura

23 September 2026 · 18:00–19:30 · Live in-person



COURSE SCHEDULE · 20 SESSIONS · 80 MINUTES EACH

Session by Session

1	02 Sep	Introduction · R setup · forecasting as a business problem ✓
2	03 Sep	Time plots, seasonal plots, EDA ✓
3	**23 Sep**	**Lags, ACF, PACF** **TODAY**
4	24 Sep	Decomposition, STL, Box-Cox
5	30 Sep	Benchmarks: Naive & Seasonal Naive
6	01 Oct	Simple exponential smoothing + Holt
7	07 Oct	Holt-Winters / ETS
8	08 Oct	ETS model selection + validation
9	15 Oct	**Intermediate Test — 20% · Checkpoint 1**
10	19 Oct	AR / MA
11	22 Oct	ARMA + stationarity
12	28 Oct	ARMA + integration + diagnostics
13	29 Oct	ARIMA
14	12 Nov	Seasonal ARIMA (SARIMA)
15	18 Nov	Automatic ARIMA + model selection
16	19 Nov	Complex seasonality / Fourier · Checkpoint 2
17	25 Nov	Integrated Forecasting Workshop
18	26 Nov	Business Case review & feedback
19	03 Dec	**Group presentations — 30%**
20	11 Dec	**Final exam in R — 40%**

TODAY'S SESSION

Measuring Memory

Lagged variables, autocorrelation, the correlogram and partial autocorrelation — how much of today is explained by yesterday.

FPP3 · Sections 2.8 – 2.9

TYPICAL CLASS STRUCTURE · 80 MINUTES

The Classroom Is the Laboratory

The same three blocks every session. The work on your case happens here, with me in the room.

~30 MIN

BLOCK 1

Concepts & R application

The methodology, its intuition and its assumptions — aimed at **what problem this technique solves** — then straight into its practical implementation in R.

You leave with code that runs.

~30 MIN

BLOCK 2

Business Case Lab

Teams apply or evaluate the methodology in the context of their own case. Finished code is not the requirement.

Output can be a plot, a methodological decision, a hypothesis, a data transformation, a fitted model, or a justified argument for why this technique does not apply.

~20 MIN

BLOCK 3

Business Case Clinic

Selected teams share their approach, findings or difficulties in about 5 minutes each.

The class discusses alternatives and gives feedback; I close the loop.

Block durations vary with the topic: technically demanding sessions take more time for explanation and R; later sessions shift progressively towards project development and feedback.

BY THE END OF TODAY YOU SHOULD BE ABLE TO

Session Objectives

1. Explain what a **lagged variable** is, build one, and connect it to the lag plots you drew last session.

2. Compute and read an **autocorrelation function**, including its significance bounds and what white noise looks like.

3. Diagnose **trend and seasonality from the correlogram alone**, and say what each shape implies for modelling.

4. Distinguish **ACF from PACF** and explain what “partial” removes and why that matters later for ARIMA.

5. **Know where your own case stands** against the checkpoint on slide 25 — Block 1 finishes tomorrow with decomposition.

BLOCK 1 · 30 MINUTES

Concept + R Application

From a picture of memory to a measurement of it.

What Is a Lagged Variable?

A lagged variable is the same series, shifted forward in time. `lag(y, k)` puts the value from *k* periods ago next to today's value, so the two can be compared row by row. Everything in this session is built on that one operation.

Month	y_t	y_{t-1}	y_{t-12}
Jan 24	1980	—	—
Feb 24	1735	1980	—
Mar 24	2620	1735	—
…	…	…	…
Jan 25	2500	2725	1980
Feb 25	2605	2500	1735

Two consequences: the first *k* rows are lost, and the lagged column is a legitimate predictor because its value is known before y_t happens.

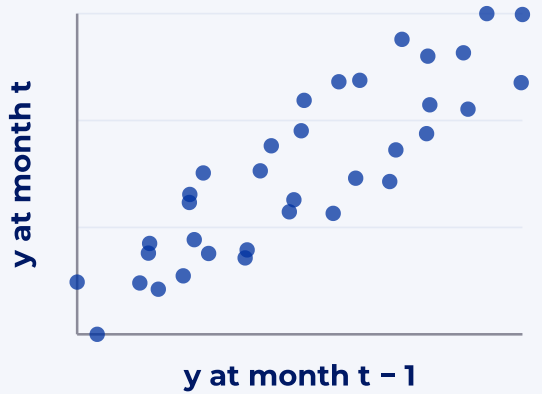
Why lags matter

- **They are the only predictors you always have.** No external data collection, no forecast-of-a-forecast problem — the past is already observed.
- **They encode the business reality.** Yesterday's queue length affects today's. Last July's ice cream sales tell you about this July.
- **They are the entire vocabulary of ARIMA.** Everything from Session 10 onwards is a statement about which lags matter and how much.
- **They convert a time series problem into a regression-shaped table** — and that is why lag plots and scatterplots look the same.

The Lag Plot Is a Scatterplot. Its Correlation Has a Name.

Plot y_t against y_{t-k} and you get an ordinary scatterplot. Compute its correlation and you get r_k — the autocorrelation at lag k. That is the whole idea; the rest is bookkeeping.

Lag 1



Moderate positive cloud: consecutive months are similar, but not by much — the seasonal swing dominates.

Lag 12



Tight and diagonal: each month closely resembles the same month a year earlier. This *is* the seasonality, expressed as a relationship.

Reading lag plots

- **Tight diagonal** → strong positive autocorrelation at that lag.
- **Shapeless cloud** → no linear relationship at that lag; the memory does not reach that far.
- **Downward diagonal** → negative autocorrelation: highs follow lows. Common at half the seasonal period.
- Drawing one scatterplot per lag does not scale. We need all of them on one chart — that is the correlogram.

The Autocorrelation Coefficient

Ordinary correlation between two variables, applied to a series and a shifted copy of itself. One difference matters: the mean and the denominator use the **whole** series, not just the overlapping rows, so the coefficients stay comparable across lags.

$$r_k = \frac{\sum_{t=k+1}^{T} (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T} (y_t - \bar{y})^2}$$

T is the length of the series, k the lag, and $\bar{y}$ the mean of the whole series. r_k always lies between −1 and 1, and $r_0 = 1$ by construction.

What to notice

- **The numerator** asks: when the series was above average k periods ago, was it above average today? Summed over every pair.
- **The denominator** is the total variation, so r_k is a proportion, not a quantity — units cancel.
- **Fewer pairs at longer lags.** At lag k you have only T − k products, so high-lag coefficients are noisier. Rule of thumb: do not read beyond T/4.
- **The collection $\{r_1, r_2, r_3, \ldots\}$ is the autocorrelation function.** Plotted, it is the correlogram.

EXAMPLE THE HAPPY COW SERIES FROM SESSION 2

The Correlogram: Every Lag on One Chart



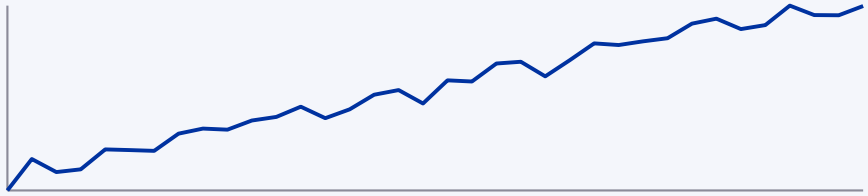
One bar per lag. Height is the correlation between the series and itself k periods earlier.

- **The shaded band** is the range in which a coefficient is indistinguishable from zero: ±1.96/√T. Bars inside it are noise.
- **Bars at 12 and 24** stand tall — the annual season. Every twelve months the series repeats.
- **Bars at 6 and 18** dip negative — half a cycle away, peaks sit opposite troughs.
- **The slow decay** of the whole pattern is the trend leaking into the autocorrelation.
- Read it as a fingerprint: the shape tells you what is in the series before you fit anything.

What a Trend Looks Like

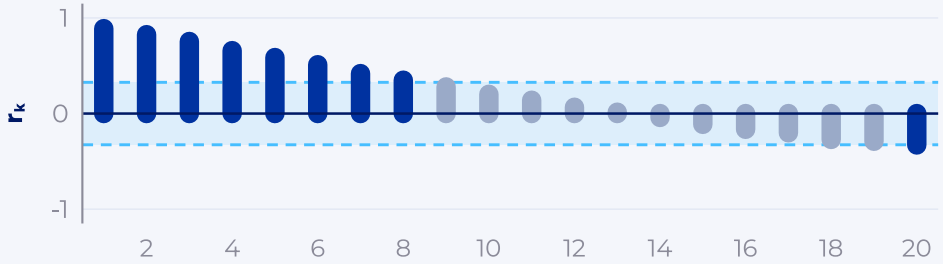
When a series trends, values close together in time are close in value — and so are values moderately far apart. The result is a correlogram that starts high and decays slowly, staying positive for many lags.

The series



Steady upward drift, no repeating shape.

Its ACF



Large at lag 1, decaying gradually, every bar positive.

Consequences

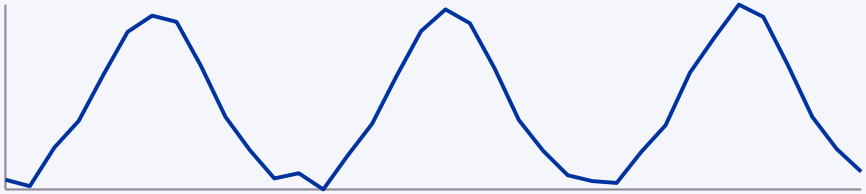
- A slowly decaying ACF is the classic signature of **non-stationarity** — the concept that opens Session 11.
- It is also why you cannot read seasonality off a trended series' ACF without care: the trend inflates every bar.
- The fix is **differencing**: model the change instead of the level. We do it properly in Block 3.

A correlogram that refuses to die is telling you the series has no fixed mean. Believe it.

What Seasonality Looks Like

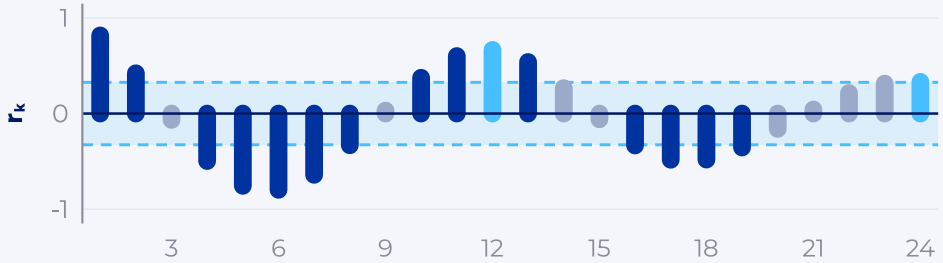
A seasonal series correlates with itself at multiples of the seasonal period, and anti-correlates half a period away. The correlogram develops a wave whose wavelength **is** the seasonal period.

The series



Stable level, twelve-month repeating shape.

Its ACF



Peaks at 12 and 24 (highlighted), troughs at 6 and 18.

How to use it

- **Do not assume the period — read it.** The lag of the first big spike names the season: 7 for daily data with a weekly cycle, 12 for monthly, 24 for hourly.
- **Multiple seasonalities show as multiple spike families**, for example at 24, 48 and 72 *and* at 168 for hourly data.
- **Spikes that fade over successive multiples** mean the seasonality is weakening — check the subseries plot.

This is the formal version of your `gg_season()` plot: same finding, now with a number and a significance threshold attached.

White Noise: the Absence of Structure

White noise is a series with no autocorrelation at any lag — nothing about the past helps predict the future. It is the null hypothesis of forecasting, and the target for your model's **residuals**.

The series



Jagged, mean-reverting, no pattern to name.

Its ACF



Every bar inside the band. Nothing to model.

The two uses

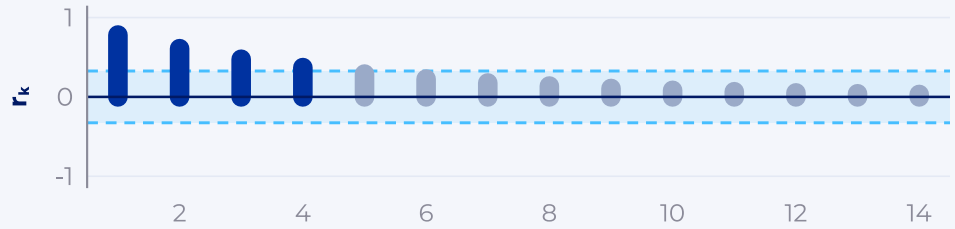
- **On the data:** if your series is white noise, no method will beat the mean. That is a finding, not a failure — report it and change the question.
- **On the residuals:** after fitting, what is left should be white noise. Structure remaining is information the model failed to capture.
- **Expect 1 bar in 20 outside the band** by chance. A pattern of spikes is evidence; a single one is not. Ljung-Box arrives in Session 5.

"Is there anything here at all?" is the first question of any engagement. The ACF answers it in one line of code.

Partial Autocorrelation: Removing the Chain

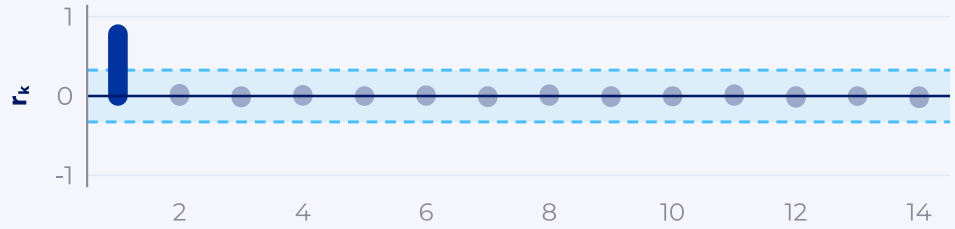
The ACF has a blind spot. If today depends on yesterday, and yesterday on the day before, then today correlates with two days ago **even if there is no direct link** — the correlation is inherited through the chain. The PACF strips that out.

Total association



ACF. Lags 2, 3 and 4 all look significant — but each only inherits the influence of lag 1.

Direct association



PACF. One spike at lag 1, then nothing: the direct memory is exactly one period long.

The intuition

- The partial autocorrelation at lag k is the coefficient on y_{t-k} in a regression that **also** includes $y_{t-1} \dots y_{t-k+1}$.
- It answers: "does lag k add anything **new**, once the shorter lags have had their say?"
- ACF asks about total association; PACF asks about direct association. Both, together, identify the model.

This is the tool that will choose your ARIMA orders in Block 3. Today, just learn to read the two charts together.

Reading the Two Together

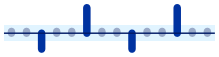
You will use this table constantly from Session 10 onwards. Today, use it to describe what your case's series is doing — not yet to name a model.

WHAT THE SERIES HAS	ACF SHOWS	PACF SHOWS	WHAT IT MEANS FOR YOU
A trend	Slow, near-linear decay; stays positive for many lags	One large spike at lag 1, then small	Non-stationary. Difference it, or model the trend explicitly.
Seasonality	Spikes at multiples of the seasonal period m	Spike at lag m, sometimes at 2m	Seasonal terms required, and seasonal naive becomes your benchmark.
Short memory	Decays quickly to zero after a few lags	Cuts off sharply after lag p	An autoregressive structure of order p — the AR in ARIMA.
Shock persistence	Cuts off sharply after lag q	Decays gradually	A moving-average structure of order q — the MA in ARIMA.
Nothing	All bars inside the band	All bars inside the band	White noise. No method beats the mean; go back to the business question.

THE ONE SLIDE TO KEEP

Signatures: What Each Process Looks Like

Not to memorise today — but from Session 10 this is how you will choose model orders.

PROCESS	ACF	PACF	HOW YOU RECOGNISE IT
AR(p) Today is a weighted sum of recent values. Order p read from the PACF.			PACF cuts off after lag p; ACF tails off.
MA(q) Today is a weighted sum of recent shocks. Order q read from the ACF.			ACF cuts off after lag q; PACF tails off.
Seasonal A repeating calendar pattern of period m.			Spike family at m, 2m, 3m in the ACF.
Random walk Each value is the previous one plus a shock. Prices behave this way.			ACF barely decays: non-stationary. Difference it, and what is left is usually white noise.
White noise No structure at any lag.			Everything inside the band. Nothing to model.

One Assumption Behind Everything Today

The ACF and PACF assume the series is **stationary** — that its mean, variance and autocorrelation structure do not change over time. Applied to a trending or drifting series, the coefficients still compute, and they still mislead.

SYMPTOM

An ACF that will not decay

Bars stay high for ten, fifteen, twenty lags. That is not a long memory — it is a series with no fixed mean, and the correlogram is measuring the trend.

TEST

Augmented Dickey-Fuller

A formal check for a unit root. A p-value above 0.05 means you cannot reject non-stationarity; below 0.05 you can proceed. It confirms what the plot suggests — it does not replace looking.

REMEDY

Difference, then look again

Model the change instead of the level: `difference(y)`. Recompute the ACF on the differenced series — the structure that remains is the structure worth modelling.

Do this in today's lab: plot the ACF of your series, then of its first difference, and put both in the log. Session 11 makes it formal.

R APPLICATION

Computing and Plotting Them

```
library(fpp3)
# lagged columns by hand
sales |> mutate(lag1 = lag(units, 1),
                lag12 = lag(units, 12))
# the coefficients as a table
sales |> ACF(units, lag_max = 24)
# the correlogram
sales |> ACF(units) |> autoplot()
# partial autocorrelation
sales |> PACF(units) |> autoplot()
```

```
# all three diagnostics at once
sales |> gg_tsdisplay(units,
                      plot_type = "partial")
# is it white noise? (Session 5)
sales |> features(units, ljung_box,
                  lag = 24)
# after differencing
sales |> ACF(difference(units)) |>
  autoplot()
```

Start with `gg_tsdisplay()`. It puts the time plot, the ACF and the PACF on one screen — the single most efficient diagnostic in the whole course.

Five Ways the ACF Misleads

The correlogram is easy to compute and easy to over-interpret. These are the mistakes that show up in student reports and in professional ones.

Reading too many lags	At lag k there are only T − k pairs. Beyond roughly T/4 the coefficients are mostly noise. With 36 months, stop looking around lag 24 — and be sceptical past 12.
Trend masquerading as memory	A trending series produces large autocorrelations at every lag. That is arithmetic, not insight. Detrend or difference before claiming the series “has memory”.
One spike is not a finding	With a 95% band, one bar in twenty crosses it by chance. Look for a pattern — a family of spikes at multiples of a period — not for individual winners.
Linear only	Autocorrelation measures **linear** dependence, exactly like the correlation of Session 2. A perfectly predictable non-linear series can show an empty ACF.
Gaps and irregular spacing	Lag k means “k periods ago”. If your tsibble has gaps or an irregular index, lag k is not a constant amount of time and the ACF is meaningless. Fix the index first.

BLOCK 2 · 30 MINUTES

Business Case Lab

Run the ACF on your own series — and finish the data-understanding phase of your engagement.

WHAT THE ACF SHOULD TELL EACH TEAM

Where to Point the Correlogram

CASE 1 · RETAIL

Happy Cow Ice Cream

Run the ACF on daily cups. Expect a spike family at lags 7, 14, 21 — the weekly cycle. There are not enough days for a lag-365 spike, which is exactly the constraint you must document.

Then: Compare the ACF before and after removing the temperature effect. What is left is the part your model must learn from the past alone.

CASE 2 · AIRLINES

Balearic Airlines

Daily booking demand should show a strong lag-7 structure. Check whether the National Day week distorts the ACF when included.

Then: Look at the PACF: if it cuts off at lag 7 while the ACF decays, the day-of-week effect is direct rather than inherited.

CASE 3 · MANUFACTURING

Clean Air Limited

Thirty-six monthly observations is thin. Compute the ACF anyway and note how few lags are trustworthy — roughly the first nine.

Then: The COVID break will dominate the correlogram. Run it with and without those months and report both.

CASE 4 · E-COMMERCE

Wayfair

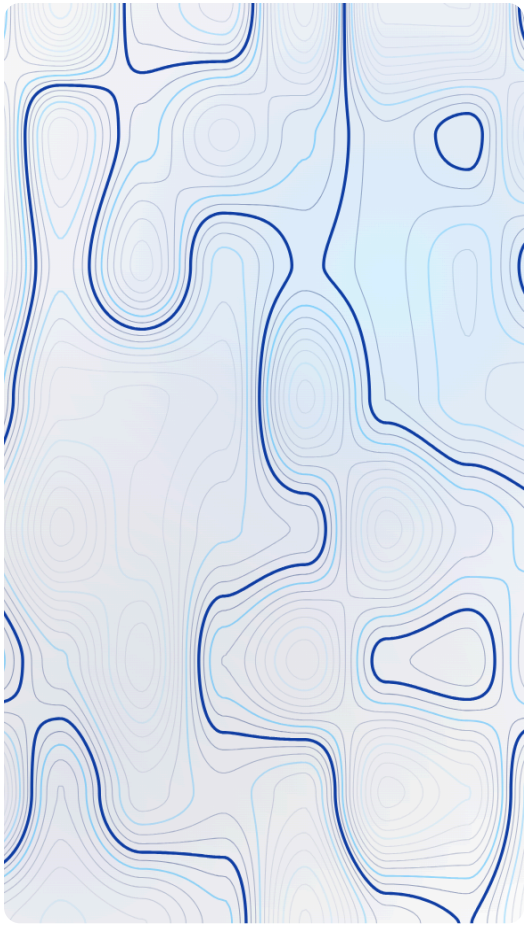
Compute the ACF at several levels of your synthetic hierarchy. Aggregate levels should show clean structure; the sparse item level often shows almost none.

Then: That contrast is the argument for hierarchical forecasting — and you can now make it with evidence.

BLOCK 3 · 20 MINUTES

Case Clinic

A few teams present. Everyone else should have something to show — you just will not know in advance whether it is you.



END OF SESSION 03

TIME SERIES ANALYSIS

Lags, Autocorrelation and the ACF

NEXT · SESSION 4 · 24 SEPTEMBER

Decomposition, transformations and STL

Tomorrow. We separate what you have just measured, and Block 1 closes.

No annex in this session. The signature cheat sheet on slide 18 is the one page worth keeping.

Angel Mora Segura · IE University

23 September 2026